Degree: BCA, Stream: BCA

Term – II, First Semester Examination, October – 2025

Subject Code: BCACC101

Subject Name: Introduction to C Programming

Full Marks: 30 Duration: 1 Hour

Part – A
Answer any 5 questions out of 6
Each question carries 2 marks (2 × 5)

1. Can we access array elements using pointers without using the [] operator? 2
2. What is the use of strlen() and strcmp()? 2
3. Define Structure in C. 2
4. What is recursive function? 2
5. What are actual and formal parameters? 2
6. What is the output of the following program:- 2

```
#include <stdio.h>
int main()
{
  int x=100;
  printf("%d\n",10+ x++);
  printf("%d\n",10+ ++x);
  return 0;
}
```

Part – B
Answer any 2 questions out of 3
Each question carries 5 marks (5 × 2)

7. Discuss advantages of array over ordinary variable. 5
8. A student writes a program using call by value to swap two numbers, but it doesn't work. Critically analyze the issue and rewrite the program using call by address to correct it. 5

9. An AI developer is implementing a large dataset of sensor readings in C. Should they use an array of pointers or a pointer to an array? Defend your choice with reasons. 5

Part – C

Answer any 1 question out of 2
Each question carries 10 marks (10 × 1)

10. The library needs a system where the librarian can add, display, search, and delete book records using a simple program. 10
Requirements: Add new books, Display all books, Search for a book by its ID, Delete a book by its ID, Exit the program.
Create a C program to implement the above.

11. A school needs a system to manage student records. The system should allow the administrator to add new students, display all students, search for a student by their ID, and delete a student by their ID. 10

Requirements: Add new students, Display all students, Search for a student by their ID, Delete a student using their ID, Exit the program.
Create a C program to implement the above.

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